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Article

Association between racialized economic segregation and stage at diagnosis for 3 screenable cancers in New York City

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Abstract

Background: Racial and economic segregation can create barriers to timely cancer diagnosis and adversely affect survival. This study examines the association between neighborhood-level segregation, measured by the neighborhood-Index of Concentration at Extremes (n-ICE), and stage at diagnosis (advanced [regional/distant] vs localized) for 3 screenable cancers in New York City.

Methods: We analyzed 98 449 incident cases (breast, 58 970; cervical, 4790; and colorectal, 34 689) using New York State Cancer Registry data (2008-2019). Census tract-level n-ICE measures of racial and/or income-based economic segregation were calculated. Age-adjusted stage-specific incidence rates and advanced-to-localized incidence rate ratios (IRRs) were measured across n-ICE quartiles.

Results: Advanced-to-localized stage IRRs were significantly higher in the most-deprived and/or non-Hispanic Black (NHB)-concentrated areas (Q1) than the most-affluent and/or most non-Hispanic White (NHW)-concentrated areas (Q4) for breast and cervical cancer (breast: $\text{n-ICE}_{\text{Income}}$, $\text{IRR}_{\text{Q1}} = 0.71$ vs $\text{IRR}_{\text{Q4}} = 0.48$; $\text{n-ICE}_{\text{NHB}}$, $\text{IRR}_{\text{Q1}} = 0.75$ vs $\text{IRR}_{\text{Q4}} = 0.53$; $\text{n-ICE}_{\text{NHB+Income}}$, $\text{IRR}_{\text{Q1}} = 0.74$ vs $\text{IRR}_{\text{Q4}} = 0.47$; cervical: $\text{n-ICE}_{\text{Income}}$, $\text{IRR}_{\text{Q1}} = 1.30$ vs $\text{IRR}_{\text{Q4}} = 0.97$; $\text{n-ICE}_{\text{NHB}}$, $\text{IRR}_{\text{Q1}} = 1.44$ vs $\text{IRR}_{\text{Q4}} = 0.99$; $\text{n-ICE}_{\text{NHB+Income}}$, $\text{IRR}_{\text{Q1}} = 1.37$ vs $\text{IRR}_{\text{Q4}} = 0.92$) (all *P*-values < .01). Hispanic concentration alone ($\text{n-ICE}_{\text{Hispanic}}$) was not associated with disparities; however, its combination with economic deprivation was significant in both cancers (breast: $\text{n-ICE}_{\text{Hispanic+Income}}$, $\text{IRR}_{\text{Q1}} = 0.70$ vs $\text{IRR}_{\text{Q4}} = 0.47$; cervical: $\text{n-ICE}_{\text{Hispanic+Income}}$, $\text{IRR}_{\text{Q1}} = 1.31$ vs $\text{IRR}_{\text{Q4}} = 0.93$) (all *P*-values < .01). All racialized-economic segregation measures ($\text{n-ICE}_{\text{NHB+Income}}$/$\text{n-ICE}_{\text{Hispanic+Income}}$) showed increasing IRRs with higher segregation for both cancers (all *P*-trend < .04). No disparities were observed for colorectal cancer.

Conclusions: Racialized-economic segregation in New York City was associated with higher advanced-stage diagnoses of breast and cervical cancer but not colorectal cancer. These findings may partially reflect both structural barriers that delay timely diagnosis and the impact of local equity-driven initiatives that broaden colorectal cancer screening access.

Introduction

Stage at diagnosis is a critical determinant of cancer survival and long-term outcomes.[1] Despite advances in early detection,[2] minoritized racial and ethnic populations are more likely to be diagnosed with advanced-stage cancers than the non-Hispanic White (NHW) population.[3,4]

Residential segregation refers to the separation of racial, ethnic, and socioeconomic groups into different neighborhoods.[5] Segregation is a fundamental mechanism through which structural inequities impact health, particularly in marginalized neighborhoods with concentrated populations of minoritized racial and ethnic groups or residents of lower socioeconomic status (SES).[5-7] Limited education, employment, housing, and economic opportunities in these segregated neighborhoods have resulted in many minoritized populations experiencing higher levels of poverty and limited access to healthcare[6], including cancer screening services, which is an important driver of disparities in cancer stage at diagnosis.[5-7]

One of the measures to quantify residential segregation is the neighborhood-Index of Concentration at Extremes (n-ICE). Unlike traditional poverty metrics, n-ICE quantifies the proportional imbalance between the concentration of affluence and deprivation within defined geographic areas and also incorporates both racial and economic dimensions, offering a robust framework for

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understanding community-level disparities in healthcare access and cancer outcomes.[8] Research using n-ICE has shown that areas with higher levels of racial and economic segregation experience poorer outcomes across various cancers (eg, breast, colorectal, liver, and lung), including higher incidence,[9-13] lower survival,[14-17] and higher mortality, particularly in predominantly non-Hispanic Black (NHB) communities. However, few studies have focused on n-ICE's impact on stage of cancer at diagnosis,[12,18,19] especially in smaller geographic areas, including at the census-tract level.[18] Evidence on disparities in stage of cancer diagnosis among Hispanic populations also remains limited,[12,18,19] despite being the second largest ethnic group in the country[20] and facing unique challenges in access to care due to cultural and linguistic barriers.[21]

In order to address these knowledge gaps, this study investigates the relationship between racial and economic segregation, as measured by census-tract level n-ICE, and stage at diagnosis for 3 screenable cancers—breast, cervical, and colorectal—in New York City. We selected these 3 screenable cancers because previous studies have shown marked disparities in stage at diagnosis by socioeconomic status and race/ethnicity for these cancers.[3,4] Moreover, New York City is an ideal setting for examining how structural inequities impact cancer disparities due to its significant racial and socioeconomic diversity[22] within a relatively small geographic area.

Methods

This study included all incident cases of first primary invasive breast, cervical, and colorectal cancers (total $n = 99\ 703$) diagnosed from 2008 through 2019 among individuals aged ≥20 years in New York City. All cases were obtained from the New York State Cancer Registry, a legally mandated, population-based cancer registry maintained by the New York State Department of Health.[23] Cases were identified using the International Classification of Diseases for Oncology, Third Edition (ICD-O-3) primary site codes. The cancer sites were classified as follows: female breast cancer (C500–C509), cervical cancer (C530–C539), and colorectal cancer (C18, C182–189, C199, C209), all excluding histology codes 9050–9055, 9140, and 9590–9992. Cases with missing data for key variables were excluded. These exclusions included cases with missing age ($n = 101$), missing sex ($n = 7$), posthumously diagnosed cases (via autopsy or death certificate; $n = 973$), and cases with low-quality geocodes ($n = 110$) or residing in census tracts with unavailable n-ICE data ($n = 63$). After exclusions, 98 449 cases were included in this study. Stage at diagnosis was identified as localized (confined to the organ of origin) or advanced (including regional and distant extension) using the Surveillance, Epidemiology, and End Results (SEER) Summary Stage, consistent with prior research.[3] While providing a clinically relevant indicator for early detection, this classification ensured sufficient statistical power, given the relatively small number of distant-stage diagnoses in some strata in this analysis. Race and ethnicity were categorized as non-Hispanic White, non-Hispanic Black, Hispanic, and any other combined (non-Hispanic Asian/Pacific Islander, Native American, and other/unknown race).

Census tract-level n-ICE measures were linked to cancer cases from the New York State Cancer Registry based on the residential census tract in the year of diagnosis. We calculated 3 n-ICE measures using the American Community Survey 5-year estimates for each year between 2008 and 2019 based on prior research[9]: (1) by income (n-ICE$_{\text{Income}}$), comparing the number of households with annual income ≤$20 000 (most-deprived) to those with income ≥$125 000 (most-affluent); (2) by race/ethnicity (n-ICE$_{\text{NHB}}$, n-ICE$_{\text{Hispanic}}$), comparing the number of non-Hispanic Black or Hispanic households (most non-Hispanic Black/Hispanic-concentrated) separately to non-Hispanic White households (most non-Hispanic White-concentrated); and (3) combining race/ethnicity and income (n-ICE$_{\text{NHB+Income}}$, n-ICE$_{\text{Hispanic+Income}}$), comparing the number of non-Hispanic Black or Hispanic households with incomes ≤$20 000 (most-deprived and most non-Hispanic Black/Hispanic-concentrated) to non-Hispanic White households with incomes ≥$125 000 (most-affluent and most non-Hispanic White-concentrated). Each n-ICE measure ranged from −1 (most-deprived and/or most non-Hispanic Black/Hispanic-concentrated) to +1 (most-affluent and/or most non-Hispanic White-concentrated). Quartiles were generated for all n-ICE measures based on their distribution in New York City, with the most-affluent and/or most non-Hispanic White-concentrated (Q4) areas set as the reference group. Census tract-level population data were obtained from the National Cancer Institute's SEER program's population estimates database by year, sex, race/ethnicity, and 19 age groups.[24]

We compared baseline characteristics by stage at diagnosis using t-tests for continuous variables and chi-square tests for categorical variables across all 3 cancers, including cases with unknown stage ($n = 5040$). For each n-ICE quartile (Q1 to Q4), we calculated overall age-adjusted stage-specific incidence rates (IRs) per 100 000 population (adjusted to the 2000 US standard population) for both localized and advanced stages of each cancer at the census tract level ($n = 93\ 409$). Additionally, we computed age-adjusted advanced-to-localized incidence rate ratios (IRRs) and 95% confidence intervals (CIs) for each n-ICE quartile. Wald tests were used to calculate P-values comparing IRRs for each quartile (Q1-Q3) with Q4 (most-affluent and/or most non-Hispanic White-concentrated). P-trend was calculated across all 4 quartiles (Q1 to Q4) using linear regression on the ordinal quartile variable to assess the trend in association. We next repeated the IRR calculations for 2 groups stratified by screening eligibility, ie, individuals who were age-eligible for cancer screening according to US Preventive Services Task Force guidelines (ages 50–74 for breast cancer, 21–65 for cervical cancer, and 50–75 for colorectal cancer),[25] and individuals who were not eligible. It should be noted that IRs and IRRs shown for quartiles of n-ICE measures represent IR and IRRs for the entire population residing in those quartiles irrespective of race/ethnicity or income. For example, non-Hispanic Black-concentrated census tracts (n-ICE$_{\text{NHB}}$) were predominantly inhabited by Black individuals, but people of other groups, including non-Hispanic White individuals, also generally resided in those tracts and were included in IRs and IRRs for that n-ICE category. All analyses were performed using SAS version 9.4 (SAS Institute, Cary, NC), with 2-sided P-values, and statistical significance was set at $P < .05$. Figures were generated using R statistical software. This study was determined as not human subjects research and exempted from review by Temple University Institutional Review Board.

Results

Study population

The study population included 58 970 breast, 4790 cervical, and 34 689 colorectal cancer cases (Table 1). On average, patients with colorectal cancer were older (mean age = 65.5 [SD = 14.6] years) compared to those with cervical cancer (53.6 [SD = 15.2] years) or breast cancer (54.4 [SD = 13.2] years). Overall, the

Table 1. Demographic and clinical characteristics of the study population by cancer type and stage at diagnosis including unknown stage, New York City, 2008-2019 ($n = 98\,449$).

	Breast cancer				Cervical cancer				Colorectal cancer			
	Localized	Advanced	Unknown	Total	Localized	Advanced	Unknown	Total	Localized	Advanced	Unknown	Total
No. (%)	35 649 (60.5)	21 523 (36.5)	1798 (3.1)	58 970	1937 (40.4)	2490 (52.0)	363 (7.6)	4790	12 148 (35.0)	19 662 (56.7)	2879 (8.3)	34 689
Age (years) Mean (SD)	60.5 (13.6)	58.1 (14.6)	64.3 (18.1)	54.4 (13.2)	49.0 (13.9)	56.6 (15.0)	57.2 (17.5)	53.6 (15.2)	65.5 (14.1)	65.1 (14.6)	68.6 (16.5)	65.5 (14.6)
Age group												
<55	12 325 (55.8)	9150 (41.5)	601 (2.7)	22 076	1314 (50.5)	1127 (43.3)	162 (6.2)	2603	2771 (34.0)	4754 (58.3)	625 (7.7)	8150
55-64	9347 (62.4)	5322 (35.5)	305 (2.0)	14 974	341 (33.0)	612 (59.2)	81 (7.8)	1034	2905 (35.7)	4630 (56.9)	599 (7.4)	8134
65-74	8179 (66.0)	3915 (31.6)	304 (2.5)	12 398	177 (27.7)	408 (64.0)	53 (8.3)	638	2977 (36.6)	4668 (57.3)	501 (6.2)	8146
75-84	4353 (63.8)	2204 (32.3)	269 (3.9)	6826	86 (22.5)	264 (68.9)	33 (8.6)	383	2406 (35.8)	3790 (56.4)	528 (7.9)	6724
≥85	1445 (53.6)	932 (34.6)	319 (11.8)	2696	19 (14.4)	79 (59.9)	34 (25.8)	132	1089 (30.8)	1820 (51.5)	626 (17.7)	3535
Sex												
Male	–	–	–	–	–	–	–	–	5954 (34.7)	9814 (57.2)	1380 (8.1)	17 148
Female	35 649 (60.5)	21 523 (36.5)	1798 (3.1)	58 970	1937 (40.4)	2490 (52.0)	363 (7.6)	4790	6194 (35.3)	9848 (56.1)	1499 (8.6)	17 541
Race/ethnicity												
Non-Hispanic White	15 246 (63.8)	7957 (33.3)	684 (2.9)	23 887	508 (43.1)	560 (47.5)	112 (9.5)	1180	4867 (35.1)	7960 (57.4)	1031 (7.4)	13 858
Non-Hispanic Black	7999 (54.0)	6375 (43.1)	432 (2.9)	14 806	470 (32.2)	881 (60.3)	111 (7.6)	1462	2940 (34.1)	4919 (57.0)	774 (9.0)	8633
Hispanic	7591 (60.0)	4710 (37.3)	344 (2.7)	12 645	611 (44.3)	691 (50.1)	78 (5.7)	1380	2636 (35.1)	4317 (57.5)	550 (7.3)	7503
Non-Hispanic-Other	4813 (63.1)	2481 (32.5)	338 (4.4)	7632	348 (45.3)	358 (46.6)	62 (8.1)	768	1705 (36.3)	2466 (52.5)	524 (11.2)	4695
n-ICE Income												
Q4 (most-affluent)	8737 (65.7)	4170 (31.3)	401 (3.0)	13 308	276 (44.7)	289 (46.8)	53 (8.6)	618	2152 (35.2)	3503 (57.4)	451 (7.4)	6106
Q3	8775 (61.3)	5152 (36.0)	391 (2.7)	14 318	412 (39.9)	535 (51.8)	85 (8.2)	1032	3003 (35.1)	4906 (57.4)	641 (7.5)	8550
Q2	7898 (59.3)	4994 (37.5)	426 (3.2)	13 318	494 (40.6)	642 (52.7)	82 (6.7)	1218	2941 (34.5)	4830 (56.7)	751 (8.8)	8522
Q1 (most-deprived)	10 239 (56.8)	7207 (40.0)	580 (3.2)	18 026	755 (39.3)	1024 (53.3)	143 (7.4)	1922	4052 (35.2)	6423 (55.8)	1036 (9.0)	11 511
n-ICE NHB												
Q4 (most NHW-concentrated)	6812 (64.2)	3489 (32.9)	308 (2.9)	10 609	295 (44.0)	317 (47.2)	59 (8.8)	671	2282 (35.5)	3654 (56.8)	499 (7.8)	6435
Q3	8616 (63.4)	4550 (33.5)	419 (3.1)	13 585	428 (45.5)	452 (48.1)	60 (6.4)	940	2830 (36.0)	4403 (56.1)	619 (7.9)	7852
Q2	8430 (61.7)	4817 (35.2)	423 (3.1)	13 670	423 (40.8)	537 (51.8)	76 (7.3)	1036	2730 (34.6)	4511 (57.2)	647 (8.2)	7888
Q1 (most NHB-concentrated)	11 791 (55.9)	8667 (41.1)	648 (3.1)	21 106	791 (36.9)	1184 (55.3)	168 (7.8)	2143	4306 (34.4)	7094 (56.7)	1114 (8.9)	12 514
n-ICE Hispanic												
Q4 (most NHW-concentrated)	1993 (60.1)	1222 (36.9)	100 (3.0)	3315	79 (37.4)	112 (53.1)	20 (9.5)	211	694 (35.3)	1097 (55.7)	178 (9.0)	1969
Q3	8655 (62.8)	4706 (34.2)	420 (3.1)	13 781	342 (40.8)	420 (50.1)	76 (9.1)	838	2523 (35.2)	4075 (56.8)	573 (8.0)	7171
Q2	11 369 (60.6)	6823 (36.4)	562 (3.0)	18 754	568 (40.5)	725 (51.8)	108 (7.7)	1401	3884 (35.0)	6275 (56.6)	931 (8.4)	11 090
Q1 (most Hispanic-concentrated)	13 632 (59.0)	8772 (37.9)	716 (3.1)	23 120	948 (40.5)	1233 (52.7)	159 (6.8)	2340	5047 (34.9)	8215 (56.8)	1197 (8.3)	14 459
n-ICE NHB + Income												
Q4 (most affluent and most NHW-concentrated)	8531 (66.3)	3983 (31.0)	353 (2.7)	12 867	247 (45.2)	244 (44.6)	56 (10.2)	547	2055 (34.8)	3417 (57.8)	436 (7.4)	5908
Q3	6421 (62.3)	3558 (34.5)	329 (3.2)	10 308	294 (44.5)	320 (48.4)	47 (7.1)	661	2224 (35.7)	3545 (56.9)	458 (7.4)	6227
Q2	7740 (61.1)	4507 (35.6)	418 (3.3)	12 665	453 (40.1)	594 (52.6)	82 (7.3)	1129	3052 (34.8)	4930 (56.2)	792 (9.0)	8774
Q1 (most-deprived and most NHB-concentrated)	12 957 (56.0)	9475 (41.0)	698 (3.0)	23 130	943 (38.4)	1332 (54.3)	178 (7.3)	2453	4817 (35.0)	7770 (56.4)	1193 (8.7)	13 780
n-ICE Hispanic + Income												
Q4 (most affluent and most NHW-concentrated)	8342 (66.5)	3865 (30.8)	343 (2.7)	12 550	240 (44.7)	241 (44.9)	56 (10.4)	537	1991 (34.8)	3312 (57.9)	414 (7.2)	5717
Q3	5940 (62.0)	3340 (34.9)	305 (3.2)	9585	253 (43.8)	282 (48.8)	43 (7.4)	578	2096 (36.0)	3306 (56.7)	429 (7.4)	5831
Q2	7627 (59.6)	4760 (37.2)	420 (3.3)	12 807	422 (38.7)	585 (53.7)	83 (7.6)	1090	2804 (34.0)	4708 (57.2)	726 (8.8)	8238
Q1 (most-deprived and most Hispanic-concentrated)	13 740 (57.2)	9558 (39.8)	730 (3.0)	24 028	1022 (39.5)	1382 (53.5)	181 (7.0)	2585	5257 (35.3)	8336 (55.9)	1310 (8.8)	14 903

Numbers in parentheses represent row %, except for age. The fourth quartile (Q4) of the n-ICE represents the most affluent and/or most non-Hispanic White-concentrated census-track; the first quartile (Q1) represents the most deprived and/or most non-Hispanic Black/Hispanic-concentrated census-track.
Abbreviations: NHB = non-Hispanic Black; NHW = non-Hispanic White; n-ICE = neighborhood-Index of Concentration at the Extremes.

proportion of cases with advanced-stage cancer was higher for colorectal cancer (56.7%), followed by cervical cancer (52.0%) and breast cancer (36.5%). Colorectal cancer had the highest proportion of unknown stage (8.3%), compared to breast cancer (3.1%) and cervical cancer (7.6%). The oldest age group (≥85 years) had significantly higher proportions of unknown-stage diagnoses across all cancer types. Census-tract income and racial/ethnic compositions across n-ICE quartiles are reported in Table S1. Of note, predominantly non-Hispanic Black and Hispanic neighborhoods frequently overlapped with low-income areas.

For both breast cancer and cervical cancer, the highest proportion of advanced-stage diagnoses was observed in the non-Hispanic Black population (breast cancer: 43.1%, cervical cancer: 60.3%), followed by Hispanic (breast cancer: 37.3%, cervical cancer: 50.1%) and non-Hispanic White (breast cancer: 33.3%, cervical cancer: 47.5%) populations. For colorectal cancer, the proportion of advanced-stage diagnosis was similar across the non-Hispanic Black (57.0%), Hispanic (57.5%), and non-Hispanic White (57.4%) populations.

By n-ICE measures, proportions of advanced-stage diagnoses for breast and cervical cancers were generally higher in the most-deprived and most non-Hispanic Black-concentrated tracts (Q1), with 6.5%-10% higher proportions compared to the most-affluent and most non-Hispanic White-concentrated tracts (Q4). The only exception was for the n-ICE$_{\text{Hispanic}}$ measure, which showed more comparable proportions across areas with different levels of racial concentration for the Hispanic population. There was little difference in the proportion of advanced-stage colorectal cancer cases across quartiles of any n-ICE measure.

Age-adjusted stage-specific incidence rates by n-ICE

The age-adjusted incidence rates (IRs) showed distinct patterns by cancer type and n-ICE measures (Table 2). For breast cancer, IRs of localized-stage cancer were significantly higher in the most-affluent and most non-Hispanic White-concentrated areas (Q4). Conversely, advanced-stage breast cancer IRs were slightly higher in the most-deprived and most non-Hispanic Black/Hispanic-concentrated areas (Q1) across the majority of n-ICE measures, except for n-ICE$_{\text{Hispanic}}$. For cervical and colorectal cancers, the IRs for both localized- and advanced-stage diagnoses were consistently higher in more deprived or non-Hispanic Black/Hispanic-concentrated areas (Q1), with one exception: colorectal cancer IRs were comparable across quartiles of n-ICE$_{\text{Hispanic}}$. Similar patterns were observed in the analysis of individuals age-eligible for cancer screening (Table S2).

Age-adjusted advanced-to-localized IRRs by n-ICE

Overall, age-adjusted IRs for advanced-stage breast cancer were consistently lower than localized-stage IRs across all ICE measures and quartiles, with IRRs ranging from 0.47 to 0.75 (Figure 1, A). For n-ICE$_{\text{Income}}$, IRRs increased from 0.48 (95% CI = 0.47 to 0.50) in the most-affluent areas (Q4) to 0.71 (95% CI = 0.69 to 0.74) in the most-deprived areas (Q1) (*P*-trend = .02) (Table S3). Although the overall trend for n-ICE$_{\text{NHB}}$ was not statistically significant (P-trend = .12), a significant difference was noted between the most non-Hispanic White-concentrated (IRR$_{\text{Q4}}$, 0.53; 95% CI = 0.50 to 0.55) and the most non-Hispanic Black-concentrated areas (IRR$_{\text{Q1}}$, 0.75; 95% CI = 0.73 to 0.77) (*P*-value < .01 for Q4 vs Q1). No statistically significant differences in IRRs were observed for n-ICE$_{\text{Hispanic}}$ (P-trend = .74). However, when race/ethnicity was combined with income, significantly increasing trends in advanced-to-localized IRRs were observed for both n-ICE$_{\text{NHB+Income}}$ (*P*-trend = .04) and n-ICE$_{\text{Hispanic+Income}}$ (*P*-trend = .01) as deprivation and non-Hispanic Black/Hispanic concentrated levels increased.

Cervical cancer had significantly higher age-adjusted IRs for advanced-stage than localized-stage in most-deprived and most non-Hispanic Black/Hispanic-concentrated areas (Q1) across all n-ICE measures, with IRRs ranging from 1.18 to 1.44 (Figure 2, A). The IRR for the most-deprived areas (Q1 n-ICE$_{\text{Income}}$ IRR, 1.30; 95% CI = 1.18 to 1.43) was significantly higher than the most-affluent areas (Q4 n-ICE$_{\text{Income}}$ IRR, 0.97; 95% CI = 0.82 to 1.15), although *P*-trend from Q4 to Q1 was .08 (Table S4). However, significant trends were observed for n-ICE$_{\text{NHB}}$ (*P*-trend = .04), n-ICE$_{\text{NHB+Income}}$ (*P*-trend = .02), and n-ICE$_{\text{Hispanic+Income}}$ (*P*-trend = .04), with IRRs increasing as deprivation and non-Hispanic Black/Hispanic concentration levels increased. Similar to breast cancer, no significant differences were observed across n-ICE$_{\text{Hispanic}}$ quartiles for cervical cancer.

For colorectal cancer, age-adjusted IRs for advanced-stage diagnoses were significantly higher than those for localized-stage across all n-ICE measures and quartiles (Figure 3, A). However, in contrast to breast and cervical cancers, no significant differences in advanced-to-localized stage IRRs were observed across quartiles of any n-ICE measure, with IRRs ranging from 1.56 to 1.68 (all *P*-values > .08 for Q4 vs Q1; *P*-trend > .17) (Table S5).

Results stratified by screening-eligibility

Among individuals who were age-eligible for cancer screening (Figures 1, B; 2, B; and 3, B), advanced-to-localized IRR patterns were generally similar to those in the total population. When compared to individuals who were not eligible for screening, advanced-to-localized IRRs were consistently lower among screening-eligible individuals by 13%–35% for breast cancer and 2%–26% for colorectal cancer across all n-ICE measures (Tables S3–S5). The difference in advanced-to-localized IRRs by screening eligibility was substantially larger for cervical cancer, but estimates were unstable due to small case counts in the non-eligible group.

Discussion

Using high-quality city-wide population-based cancer registry data, we found significant associations between residential economic and/or racial segregation and stage at diagnosis for breast and cervical cancer but not for colorectal cancer in New York City. Women with breast and cervical cancers consistently were more likely to be advanced stage in economically deprived and racially/ethnically segregated neighborhoods, except for the Hispanic population concentration measure alone. These results highlight that neighborhood level of economic or racial/ethnic segregation may contribute to disparities in cancer stage at diagnosis, underscoring the importance of equity-focused interventions that address structural barriers to promote earlier detection in marginalized communities.

We found that the disparities in breast and cervical cancer stage at diagnosis in non-Hispanic Black population-concentrated areas were further exacerbated by economic deprivation. Although many predominantly non-Hispanic Black communities demonstrate strong social cohesion, community engagement, and cultural assets,[26,27] these factors alone may be insufficient to overcome longstanding structural barriers in racially segregated, economically under-resourced settings. In

Table 2. Age-adjusted incidence rates/100 000 by quartile of residential racial and economic segregation measured by the neighborhood-Index of Concentration at the Extremes (n-ICE) for breast, cervical, and colorectal cancer in New York City, 2008-2019.

	Breast cancer ($n = 57\ 172$)		Cervical cancer ($n = 4427$)		Colorectal cancer ($n = 31\ 810$)	
	Localized	Advanced	Localized	Advanced	Localized	Advanced
Overall	85.5 (84.6-86.4)	52.5 (51.8-53.2)	5.0 (4.8-5.3)	6.1 (5.9-6.4)	15.8 (15.5-16.1)	25.7 (25.3-26.0)
n-ICE Income						
Q4 (most-affluent)	108.2 (105.9-110.6)	52.5 (50.8-54.1)	3.7 (3.2-4.1)	3.6 (3.1-4.0)	13.6 (13.0-14.1)	22.2 (21.4-22.9)
Q3	87.0 (85.1-88.9)	52.3 (50.8-53.7)	4.5 (4.1-5.0)	5.5 (5.0-5.9)	15.8 (15.2-16.3)	25.9 (25.2-26.6)
Q2	79.8 (78.0-81.6)	51.3 (49.8-52.7)	5.4 (4.9-5.8)	6.7 (6.1-7.2)	16.1 (15.5-16.7)	26.6 (25.8-27.4)
Q1 (most-deprived)	75.4 (74.0-76.9)	53.9 (52.6-55.1)	6.0 (5.5-6.4)	7.7 (7.2-8.2)	17.0 (16.5-17.6)	27.2 (26.5-27.9)
n-ICE NHB						
Q4 (most NHW-concentrated)	92.9 (90.6-95.1)	48.8 (47.1-50.4)	4.7 (4.2-5.2)	4.7 (4.1-5.2)	15.6 (14.9-16.2)	25.1 (24.3-26.0)
Q3	94.2 (92.2-96.3)	50.4 (48.9-51.9)	5.0 (4.5-5.5)	5.0 (4.5-5.5)	15.8 (15.2-16.4)	24.6 (23.9-25.4)
Q2	86.0 (84.2-87.9)	50.0 (48.6-51.5)	4.7 (4.3-5.2)	5.6 (5.2-6.1)	14.8 (14.2-15.3)	24.5 (23.8-25.2)
Q1 (most NHB-concentrated)	77.0 (75.6-78.4)	57.6 (56.3-58.8)	5.5 (5.1-5.8)	7.8 (7.4-8.3)	16.5 (16.0-17.0)	27.4 (26.8-28.1)
n-ICE Hispanic						
Q4 (most NHW-concentrated)	88.9 (84.8-92.9)	56.7 (53.4-60.1)	4.3 (3.3-5.3)	5.2 (4.2-6.2)	16.2 (15.0-17.5)	25.6 (24.0-27.2)
Q3	96.2 (94.1-98.3)	53.7 (52.2-55.3)	4.2 (3.8-4.7)	4.8 (4.3-5.2)	14.6 (14.0-15.2)	23.7 (23.0-24.5)
Q2	89.4 (87.7-91.1)	54.6 (53.3-55.9)	4.9 (4.5-5.3)	5.8 (5.3-6.2)	16.1 (15.6-16.7)	26.2 (25.5-26.9)
Q1 (most Hispanic-concentrated)	77.1 (75.8-78.4)	50.2 (49.1-51.2)	5.6 (5.2-6.0)	7.1 (6.7-7.5)	16.1 (15.7-16.6)	26.3 (25.8-26.9)
n-ICE NHB + Income						
Q4 (most affluent and most NHW-concentrated)	110.0 (107.6-112.5)	52.2 (50.5-53.9)	3.4 (3.0-3.9)	3.2 (2.8-3.6)	13.4 (12.8-13.9)	22.4 (21.6-23.2)
Q3	90.1 (87.8-92.4)	51.1 (49.4-52.8)	4.8 (4.2-5.3)	4.7 (4.2-5.2)	15.6 (14.9-16.2)	25.0 (24.2-25.9)
Q2	81.0 (79.1-82.8)	47.9 (46.5-49.4)	5.2 (4.7-5.7)	6.4 (5.9-6.9)	16.7 (16.1-17.4)	27.1 (26.3-27.9)
Q1 (most-deprived and most NHB-concentrated)	75.8 (74.5-77.1)	56.3 (55.1-57.4)	5.8 (5.4-6.2)	7.9 (7.5-8.4)	16.5 (16.0-16.9)	26.9 (26.2-27.5)
n-ICE Hispanic + Income						
Q4 (most affluent and most NHW-concentrated)	111.1 (108.6-113.6)	52.3 (50.6-54.0)	3.5 (3.0-3.9)	3.2 (2.8-3.6)	13.3 (12.7-13.9)	22.4 (21.6-23.2)
Q3	89.2 (86.9-91.6)	51.3 (49.5-53.1)	4.3 (3.8-4.9)	4.4 (3.9-4.9)	15.8 (15.1-16.4)	25.1 (24.3-26.0)
Q2	83.1 (81.2-85.0)	53.0 (51.5-54.6)	5.1 (4.6-5.6)	6.5 (6.0-7.1)	16.2 (15.6-16.8)	27.3 (26.5-28.1)
Q1 (most-deprived and most Hispanic-concentrated)	75.5 (74.2-76.8)	53.2 (52.1-54.3)	5.9 (5.5-6.3)	7.7 (7.3-8.2)	16.6 (16.2-17.1)	26.6 (26.1-27.2)

Numbers in parentheses represent 95% confidence intervals (CIs). The fourth quartile (Q4) of the n-ICE represents the most affluent and/or most non-Hispanic White-concentrated census-track; the first quartile (Q1) represents the most deprived and/or most non-Hispanic Black/Hispanic-concentrated census-track.
Abbreviations: NHB = non-Hispanic Black; NHW = non-Hispanic White.

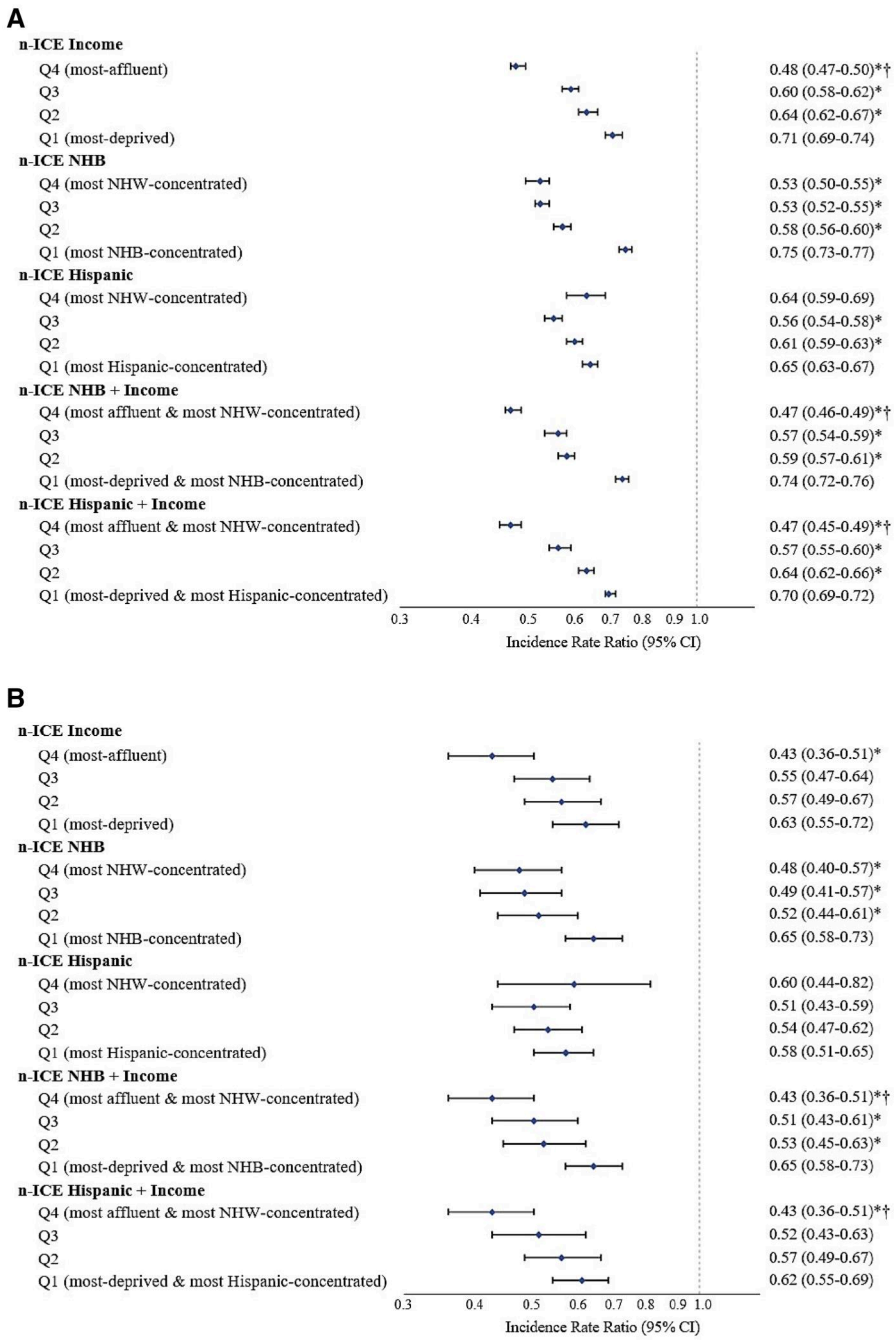

Figure 1. Breast cancer advanced-to-localized incidence rate ratio by the neighborhood-Index of Concentration at the Extremes (n-ICE), New York City, 2008-2019. (A) All age (≥20 years); (B) Screening-eligible (50-74 years). Bars in the graph and numbers in parentheses represent 95% confidence intervals (CIs). *P-value for the difference between Q1-Q3 and Q4 (the most-affluent and/or most non-Hispanic White-concentrated) was significant at <.05. †P for trend across 4 quartiles (Q4-Q1) was significant at <.05.

particular, non-Hispanic Black residents, especially those without insurance, may be disproportionately excluded from routine screening and on-time diagnoses due to financial barriers.[28] Even insured Black individuals may experience financial hardships, including transportation costs, unpaid time off work, and high out-of-pocket costs of diagnostic follow-ups, which can contribute to advanced-stage diagnoses.[6,29] These personal financial barriers stem from structural inequities shaped by historical policies like redlining, which systematically excluded Black communities from economic opportunities and long-term wealth accumulation,[12] as indicated by frequent overlapping of predominantly non-Hispanic Black and low-income neighborhoods (Table S1). These structural inequities can also influence health insurance coverage patterns, with economic disadvantage limiting access to employer-sponsored insurance and other affordable coverage options in predominantly non-Hispanic Black neighborhoods.[6] For example, in New York City, Black households have a median annual income of $58 011—less than half that of White households ($110 890)—and their uninsurance rate is more than double (6.3% vs 3.6% in 2019-2021).[30] Of note, our results showed higher incidence rates of localized-stage breast cancer in more affluent census tracts, whereas

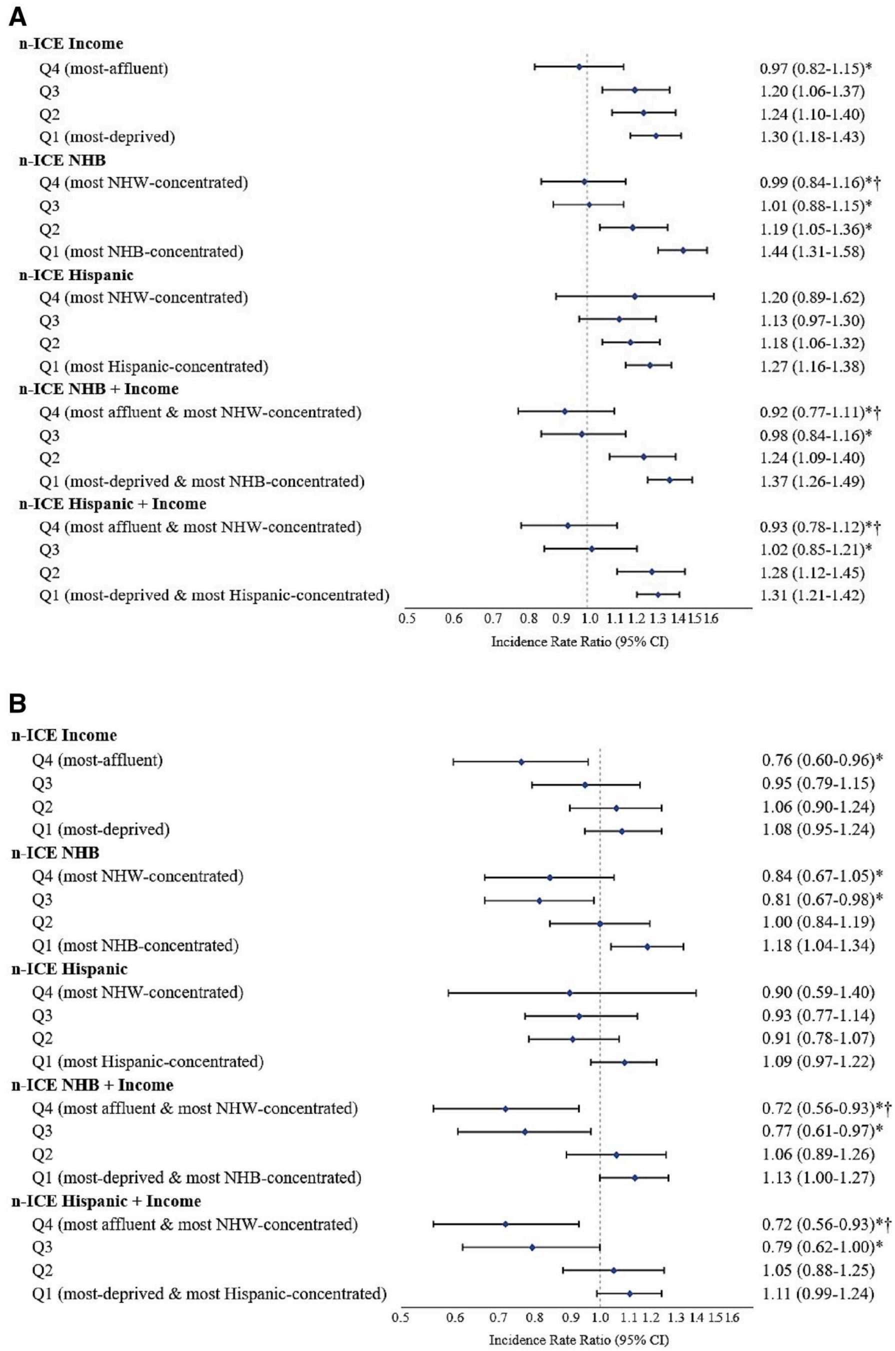

Figure 2. Cervical cancer advanced-to-localized incidence rate ratio by the neighborhood-Index of Concentration at the Extremes (ICE), New York City, 2008-2019. (A) All age (≥20 years); (B) Screening-eligible (21-65 years). Bars in the graph and numbers in parentheses represent 95% confidence intervals (CIs). *P-value for the difference between Q1-Q3 and Q4 (the most-affluent and/or most non-Hispanic White-concentrated) was significant at <.05. †P for trend across 4 quartiles (Q4-Q1) was significant at <.05.

advanced-stage cancer rates were comparable across n-ICE$_{\text{Income}}$ categories, perhaps in part reflecting higher rates of overdiagnosis (eg, detection of indolent localized tumors) in populations with greater screening access.[31]

Systemic disinvestment in predominantly non-Hispanic Black neighborhoods has further restricted access to quality care by undermining healthcare infrastructure and limiting care options.[6,32,33] Segregated neighborhoods frequently face a shortage of Medicaid-accepting providers, limiting access to screening and diagnostic services and diminishing the potential benefits of expanded insurance coverage.[34] Insufficient investment in healthcare facilities that largely serve this population (eg, outdated diagnostic equipment) and inadequate provider training can result in long wait times for screening, delays in follow-up care, and lower care quality, leading to late-stage diagnoses.[35,36] Moreover, the chronic underfunding of safety-net hospitals in New York City[37] has resulted in hospital closures that disproportionately impact people of lower income and racially minoritized groups.[38] These closures, combined with the inequitable allocation of resources, such as New York's Indigent Care Pool—where funds are disproportionately directed to wealthier hospitals—have significantly limited the capacity of safety-net hospitals to

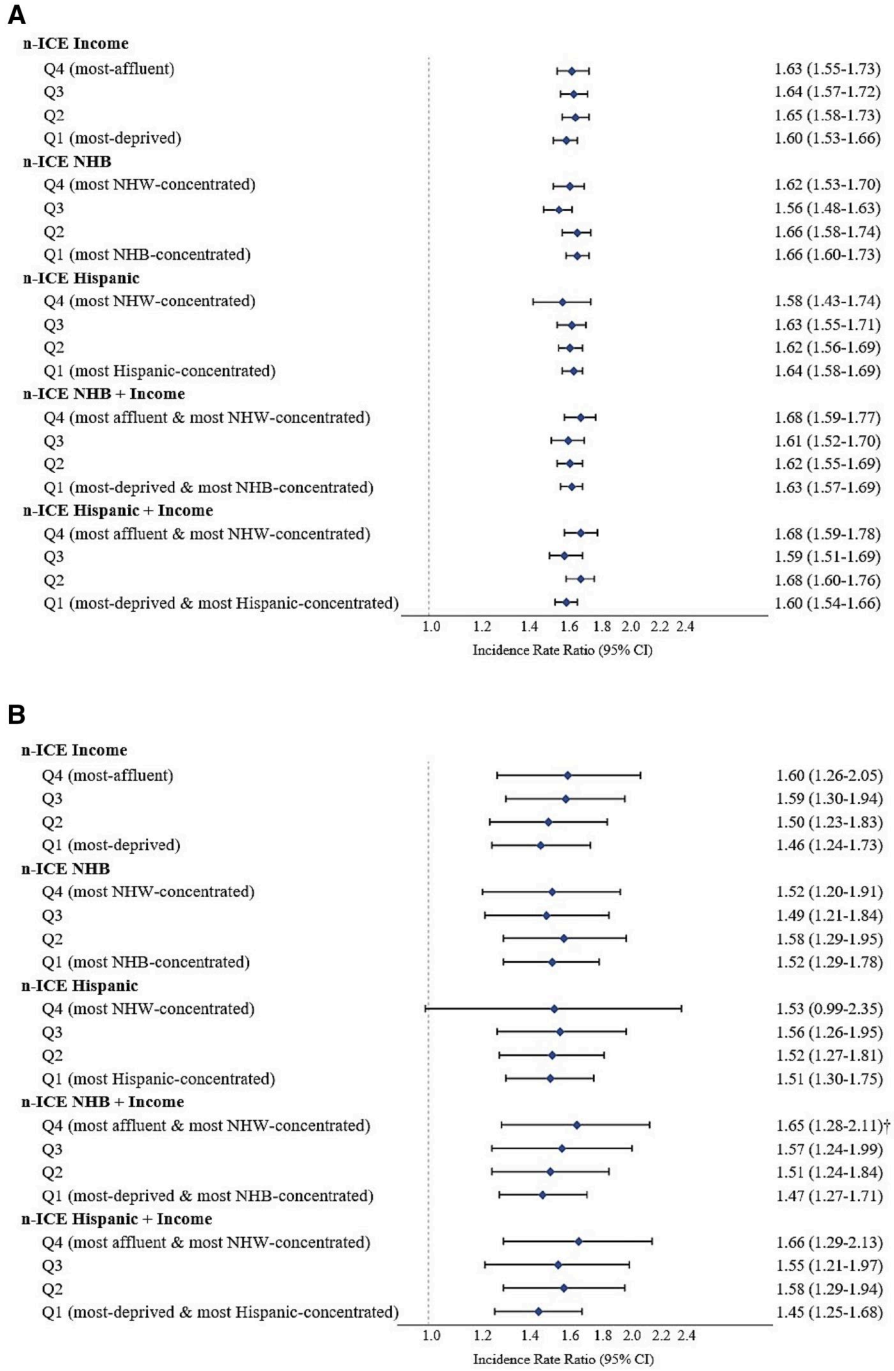

Figure 3. Colorectal cancer advanced-to-localized incidence rate ratio by the neighborhood-Index of Concentration at the Extremes (n-ICE), New York City, 2008-2019. (A) All age (≥20 years); (B) Screening-eligible (50-74 years). Bars in the graph and numbers in parentheses represent 95% confidence intervals (CIs). *P-value for the difference between Q1-Q3 and Q4 (the most-affluent and/or most non-Hispanic White-concentrated) was significant at <.05. †P for trend across 4 quartiles (Q4-Q1) was significant at <.05.

deliver timely and high-quality cancer screening and diagnosis services.[37,38] Furthermore, programs that are implemented to assist persons of lower income with access to screening, such as the Cancer Services Program and National Breast and Cervical Cancer Early Detection Program, serve a relatively small proportion of the eligible population (eg, breast cancer: 15.0% in 2018-2019; cervical cancer: 5.6% in 2018-2020), with state-by-state variation (breast cancer: 12.3%-27.7% in 2018-2019; cervical cancer: 3.9%-14.7% in 2018-2020).[39] Targeted multilevel policies, including expanding affordable healthcare coverage, strengthening healthcare infrastructure, and leveraging existing community strengths, are critical to reducing barriers to care and achieving equitable staging outcomes in breast and cervical cancer.

Hispanic population concentration alone was not significantly associated with advanced-stage diagnoses of breast and cervical cancers, suggesting that supportive characteristics within Hispanic neighborhoods may mitigate adverse health effects typically linked to racial segregation. Unlike non-Hispanic Black population concentrated neighborhoods, which have been shaped by historical housing and economic policies such as redlining,[12] Hispanic residential clustering in New York City often forms by community choice, aligning with the ethnic density perspective.[40] These culturally concordant neighborhoods can foster social cohesion and robust community networks,[41,42] which may provide protective effects through culturally appropriate health programs,[43] psychosocial support,[44] and community engagement[41,43] that promote health-seeking behaviors and facilitate early-stage diagnoses.[41-44] Our findings are consistent with prior studies indicating that Hispanic people living in residential ethnic enclaves may benefit from social resources, including mutual support and culturally sensitive care, which may enhance health outcomes regardless of poverty levels.[45]

Among Hispanics, the association of racial and economic segregations with diagnosis of advanced-stage breast and cervical cancers was confined to those residing in areas with high Hispanic population concentration and economic deprivation, suggesting that economic segregation plays a stronger role in determining disparities in stage at diagnosis than Hispanic concentration alone. Economically disadvantaged Hispanic neighborhoods face structural barriers similar to those in low-income non-Hispanic Black population-concentrated areas,[14] which can diminish the potential protective effects associated with high concentrations of Hispanic residents. These barriers are compounded by language isolation and immigration-related challenges, which further restrict health-seeking behaviors and impede navigation of the healthcare system.[40] In New York City, 72% of Hispanic immigrants have limited English proficiency,[46] and nearly 19% live below 200% of the federal poverty level,[47] confining many Spanish-speaking Hispanic residents to under-resourced neighborhoods.[48,49] For undocumented Hispanic immigrants, who comprise 53% of the city's undocumented population,[46] these challenges are further exacerbated by legal barriers, such as ineligibility for public benefits and fear of seeking care due to immigration status.[50] Although some Hispanic healthcare providers prefer working in language-concordant communities,[51,52] the overall supply of healthcare providers remains insufficient in these neighborhoods, which continue to experience shortages of primary care physicians and inadequate healthcare infrastructure,[53] leaving many residents underserved. Previous studies have reported that these challenges—regardless of individuals' nativity[54,55] and immigration status[56]—are pervasive in economically segregated Hispanic communities across the country.

Our findings on colorectal cancer differ from those of Scally et al.,[18] who reported an association between non-Hispanic Black racialized economic segregation and advanced-stage colorectal cancer diagnoses using pooled data from 18 SEER registry states. The absence of significant colorectal cancer disparities in stage at diagnosis across all residential segregation levels in New York City in our study could reflect multiple factors, including the impact of statewide and local initiatives aimed at improving colorectal cancer screening,[57] such as the Citywide Colon Cancer Control Coalition (C5) in New York City since 2003,[57] and the statewide Cancer Services Program since 2007.[58] These efforts led to substantial increases in colonoscopy screening rates—from 35% to 72% among non-Hispanic Black individuals, from 38% to 69% among Hispanic individuals, and from 48% to 69% among non-Hispanic White individuals between 2003 and 2018 in New York City,[59] and from 49.6% to 76.6% statewide between 2003[60] and 2020.[61] In addition, our colorectal cancer population had a mean age of 65.5 years (vs 54 years for breast and cervical cancers), suggesting that many were covered by Medicare, which likely reduced individual-level financial barriers and facilitated timely screening and diagnosis.

Despite these advances in colorectal cancer screening utilization and equity, Black-White mortality disparities persist in New York City.[59] Although mortality rates declined significantly from 2001 to 2019—from 34.7 to 19.2 per 100 000 among non-Hispanic Black individuals and from 28.9 to 14.8 per 100 000 among non-Hispanic White individuals—the mortality gap between these groups remained largely unchanged.[62] In many segregated neighborhoods, structural barriers such as lack of health insurance, limited resources to afford timely and guideline-concordant treatment, provider shortages in underserved areas,[63] fragmented care delivery systems,[34] and delays in specialist referrals[64] can substantially undermine the potential benefits of early detection. In contrast, Delaware has substantially narrowed Black-White mortality disparities over the same period, with non-Hispanic Black mortality declining from 31.2 to 15.1 per 100 000 and non-Hispanic White mortality decreasing from 19.5 to 13.3 per 100 000 from 2001 to 2019.[65,66] The Delaware Cancer Consortium, established in 2002, facilitates access to colorectal cancer screening, treatment, and patient navigation, complemented by culturally tailored outreach and other barrier-reduction strategies.[65] This progress demonstrates how comprehensive, coordinated interventions targeting multiple structural barriers can effectively mitigate cancer outcome disparities in segregated communities. It should be noted that New York State's Medicaid Cancer Treatment Program also provides assistance to receive colorectal cancer treatment, but with much stricter eligibility requirements compared to the Delaware Cancer Consortium programs in terms of income, age, immigration status, insurance status, and renewal requirements. For example, New York limits eligibility to individuals aged < 65 years with incomes below 250% of the FPL,[67] whereas Delaware extends eligibility to individuals of any age with incomes up to 650% of the FPL.[68]

In stratified analyses by cancer screening eligibility, disparities in stage at diagnosis were relatively smaller among screening-eligible than non-eligible populations, which may largely reflect the impact of access to screening. Screening-eligible individuals who have health insurance can generally receive cancer screening with no cost sharing, whereas for non-eligible individuals, out-of-pocket costs of early detection services can be an important barrier, especially for those with lower incomes[69] who are more likely to reside in segregated neighborhoods. Despite smaller disparities in stage at diagnosis among screening-eligible individuals, disparities for breast and cervical cancer persisted, underscoring the need for more concerted actions to increase access to and utilization of guideline-recommended cancer screening for eligible individuals in under-resourced populations, as well as early detection services for screening non-eligible individuals.

This study is the first population-based analysis at the census-tract level to examine disparities in stage of cancer at diagnosis across 3 screenable cancers within the context of racialized economic segregation. By simultaneously analyzing breast, cervical, and colorectal cancer characteristics, the study highlights contrasts in how structural inequities and efforts to mitigate those inequities impact different cancer types, emphasizing the need for targeted, cancer-specific interventions. Moreover, the inclusion of New York City's large Hispanic population enabled a detailed examination of Hispanic segregation, contributing valuable insights into an often-underrepresented area of research. The use of n-ICE measures based on both income and race/ethnicity also allowed for an examination of neighborhood-level inequities by capturing the intersections of race and economic status.

However, this study has limitations. The exclusion of cases with missing data may have introduced bias, potentially underestimating the disparities in certain groups, although the proportion of individuals with missing data in this study was relatively small. The registry does not collect data on key variables such as patient-provider communication, social support, health literacy, and whether cancers were detected through screening or symptomatic presentation, limiting the ability to explore mechanisms driving disparities. Thus, we cannot directly attribute the observed disparities to differences in screening uptake, adherence, or quality of follow-up care across segregation levels. Our study also did not account for potential biological differences in tumor characteristics across racial/ethnic groups, such as the higher proportion of triple-negative breast cancer among non-Hispanic Black women, a subtype that often presents at more advanced stages.[70] While such biological factors may contribute to observed disparities in stage at diagnosis, they likely explain a relatively small portion of these differences as triple-negative breast cancer accounts for about 8%-15% of all breast cancer cases.[70] Additionally, due to small numbers or insufficient data we were unable to examine disparities among other racial/ethnic groups, such as Asian or Native American populations, by birthplace, and among subgroups of Hispanic (eg, Puerto Rican, Central American) and non-Hispanic Black (eg, Afro-Caribbean) populations, which may have obscured the unique challenges experienced by those subgroups. We also did not consider lung cancer in this analysis because the nationwide screening uptake was low.[71] Finally, while New York City's diversity and robust healthcare system provide an ideal setting for studying segregation, the findings may not be generalizable to other areas with different healthcare systems, local public-health initiatives, access to care, demographic compositions, and segregation patterns. The cross-sectional design also limits causal inferences about the relationship between segregation and cancer stage at diagnosis.

Conclusion

This study highlights significant disparities in breast and cervical cancer stage at diagnosis linked to residential sociodemographic segregation in New York City, with variations by cancer type and segregation patterns. The absence of colorectal cancer staging disparities may reflect the combined impact of broader healthcare coverage and sustained, equity-driven local screening initiatives. Future research should assess segregation's impact across more diverse populations and geographic regions and incorporate social determinants of health to guide effective interventions.

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Author contributions

Qinran Liu (Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Resources, Software, Validation, Visualization, Writing—original draft), Daniel Wiese (Conceptualization, Data curation, Funding acquisition, Investigation, Methodology, Resources, Supervision, Validation, Visualization, Writing—review & editing), Paulo S. Pinheiro (Writing—review & editing), Jordan Baeker Bispo (Data curation, Writing—review & editing), Margaret Gates Kuliszewski (Data curation, Resources, Writing—review & editing), Tabassum Z. Insaf (Data curation, Resources, Validation, Writing—review & editing), Kevin A. Henry (Writing—review & editing), Ahmedin Jemal (Funding acquisition, Resources, Writing—review & editing), and Farhad Islami (Conceptualization, Data curation, Funding acquisition, Investigation, Methodology, Project administration, Resources, Supervision, Validation, Writing—review & editing)

Supplementary material

Supplementary material is available at *JNCI: Journal of the National Cancer Institute* online.

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Conflicts of interest

Qinran Liu, Daniel Wiese, Jordan Baeker Bispo, Ahmedin Jemal, and Farhad Islami are employed by the American Cancer Society, which receives grants from private and corporate foundations, including foundations associated with companies in the health sector for research outside the submitted work; these authors have nothing else to disclose. Paulo S. Pinheiro, Margaret Gates Kuliszewski, Tabassum Z. Insaf, and Kevin A. Henry have nothing to disclose.

Data availability

This study was determined as not human subjects research and exempted from review by Temple University Institutional Review Board. The data generated in this study are available upon request from the corresponding author, subject to compliance with the legal authorization of the New York State Cancer Registry.

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